

# AI-Enhanced Intrusion Detection System

## Final Project Documentation

SmartInternz Externship Program — 2026

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**Submitted By:**

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Domain: Cybersecurity & Machine Learning

Technology: Python Flask | scikit-learn | Bootstrap 5

Dataset: NSL-KDD Network Intrusion Dataset

# 1. INTRODUCTION

## 1.1 Project Overview

The AI-Enhanced Intrusion Detection System (IDS) is a machine learning-powered cybersecurity web application designed to detect and classify network intrusions in real time. The system leverages a Random Forest Classifier trained on NSL-KDD dataset features to identify five distinct categories of network traffic: Normal, DoS (Denial of Service), Probe, R2L (Remote to Local), and U2R (User to Root) attacks.

Built using Python Flask for the backend, Bootstrap 5 for the responsive frontend, and scikit-learn for the ML pipeline, the application provides a comprehensive platform that includes a live dashboard, a network traffic prediction form, an alert log system, and a REST API endpoint for seamless programmatic integration.

## 1.2 Purpose

The primary purpose of this project is to bridge the gap between academic machine learning research and practical cybersecurity tooling. Traditional intrusion detection systems rely on rule-based methods that struggle to adapt to evolving attack patterns. This project demonstrates how modern machine learning techniques can:

- Automatically classify network traffic without manual rule definition
- Provide real-time detection with confidence scoring
- Offer an accessible web interface suitable for non-technical security analysts
- Expose a REST API enabling integration into larger security infrastructure
- Log and visualize intrusion events for forensic and audit purposes

## 2. LITERATURE SURVEY

### 2.1 Existing Problem

Network security remains one of the most critical challenges in information technology. Existing intrusion detection systems (IDS) face several fundamental limitations:

- **Rule-Based Rigidity:** Traditional signature-based IDS tools such as Snort require constant manual updates to their rule sets and are unable to detect novel or zero-day attacks.
- **High False Positive Rates:** Anomaly-based systems often produce excessive false alarms, leading to alert fatigue among security personnel.
- **Scalability Issues:** As network traffic volumes grow exponentially, conventional systems struggle to maintain real-time performance.
- **Limited Adaptability:** Static models cannot adapt to evolving attack vectors without complete retraining cycles.
- **Lack of Explainability:** Many deep learning approaches function as black boxes, providing predictions without interpretable reasoning.

### 2.2 References

| # | Author(s)              | Title / Contribution                                                            | Year |
|---|------------------------|---------------------------------------------------------------------------------|------|
| 1 | Tavallaee et al.       | NSL-KDD: An improved dataset for network intrusion detection systems evaluation | 2009 |
| 2 | Breiman, L.            | Random Forests — foundational paper on ensemble decision tree classifier        | 2001 |
| 3 | Pedregosa et al.       | Scikit-learn: Machine Learning in Python — ML library used for model training   | 2011 |
| 4 | Palvinder Singh et al. | Network Intrusion Detection using Machine Learning approaches on KDD dataset    | 2019 |
| 5 | Vinayakumar et al.     | Deep Learning Approach for Intelligent Intrusion Detection System               | 2019 |

### 2.3 Problem Statement Definition

Design and implement an intelligent, web-accessible intrusion detection system that uses machine learning to automatically classify incoming network traffic as benign or malicious, identify the specific attack category, provide real-time confidence-scored alerts, and expose programmatic integration points — all without requiring manual signature updates or deep security expertise from the end user.

### 3. IDEATION & PROPOSED SOLUTION

#### 3.1 Empathy Map Canvas

| Dimension | Insights                                                                                                                   |
|-----------|----------------------------------------------------------------------------------------------------------------------------|
| SAYS      | "I need to monitor my network without being a security expert."   "Alert me only when something is genuinely suspicious."  |
| THINKS    | "Manual rule updates are tedious."   "False positives waste my time and erode trust in the system."                        |
| DOES      | Reviews alert logs daily   Manually investigates suspicious IPs   Uses packet capture tools like Wireshark                 |
| FEELS     | Overwhelmed by the volume of network events   Anxious about undetected intrusions   Frustrated by opaque security tooling  |
| PAINS     | No easy way to classify attacks in real time   Existing tools too complex or expensive   No confidence score on detections |
| GAINS     | Real-time web dashboard   ML-powered auto-classification   REST API for automation   Confidence scores per detection       |

#### 3.2 Ideation & Brainstorming

The following ideas were explored and evaluated during the design phase:

| Idea                             | Evaluation                                                           | Decision                      |
|----------------------------------|----------------------------------------------------------------------|-------------------------------|
| Deep Neural Network (DNN)        | High accuracy but requires large data volume and is a black box      | Rejected — less interpretable |
| Support Vector Machine           | Good binary classification but limited multi-class support           | Rejected — too narrow         |
| Random Forest Classifier         | High accuracy, handles multi-class, interpretable feature importance | Selected                      |
| Real-time packet capture (scapy) | Adds live monitoring but increases complexity significantly          | Future scope                  |
| Flask Web Interface              | Lightweight, easy to deploy, Python-native integration with ML       | Selected                      |
| REST API Endpoint                | Enables programmatic integration with SOC tools and dashboards       | Selected                      |

## 4. REQUIREMENT ANALYSIS

### 4.1 Functional Requirements

| ID    | Requirement                                                                   | Priority |
|-------|-------------------------------------------------------------------------------|----------|
| FR-01 | System shall accept 10 network traffic features as input via web form         | High     |
| FR-02 | System shall classify traffic into 5 categories: Normal, DoS, Probe, R2L, U2R | High     |
| FR-03 | System shall display prediction result with confidence score                  | High     |
| FR-04 | System shall log all predictions with timestamp, label, and confidence        | High     |
| FR-05 | System shall provide a live dashboard with attack distribution chart          | Medium   |
| FR-06 | System shall expose POST /api/predict REST endpoint accepting JSON input      | High     |
| FR-07 | System shall include demo simulation buttons for each attack category         | Medium   |
| FR-08 | System shall display all historical alerts in a sortable table                | Medium   |

### 4.2 Non-Functional Requirements

| ID     | Requirement                                                            | Metric                                    |
|--------|------------------------------------------------------------------------|-------------------------------------------|
| NFR-01 | Performance: Prediction latency must be acceptably low                 | < 500ms per prediction                    |
| NFR-02 | Accuracy: Model classification accuracy on test set                    | > 95% accuracy                            |
| NFR-03 | Usability: Interface shall be responsive across device sizes           | Bootstrap 5 responsive grid               |
| NFR-04 | Reliability: Application shall remain stable under continuous requests | No crash in 1,000 sequential requests     |
| NFR-05 | Maintainability: Model can be retrained by replacing .pkl files        | Decoupled model storage                   |
| NFR-06 | Portability: Application shall run on any Python 3.8+ environment      | Cross-platform via requirements.txt       |
| NFR-07 | Security: API responses shall not expose internal model architecture   | Responses contain only label + confidence |

## 5. PROJECT DESIGN

### 5.1 Data Flow Diagrams & User Stories

#### Data Flow Diagram

**Level 0 — Context Diagram:**

User Browser → [Flask Web Application] → ML Prediction Engine → Alert Store

**Level 1 — System Data Flow:**

| Step | Process         | Data In                                     | Data Out                               |
|------|-----------------|---------------------------------------------|----------------------------------------|
| 1    | Feature Input   | 10 raw network feature values from form/API | Raw feature array                      |
| 2    | Preprocessing   | Raw feature array                           | Scaled feature vector (StandardScaler) |
| 3    | Model Inference | Scaled feature vector                       | Class label + probability array        |
| 4    | Alert Logging   | Label, confidence, timestamp                | Stored alert record                    |
| 5    | Response        | Alert record + prediction                   | JSON response / HTML result card       |

#### User Stories

| ID    | As a...                | I want to...                                   | So that...                                                |
|-------|------------------------|------------------------------------------------|-----------------------------------------------------------|
| US-01 | Security Analyst       | Submit network traffic features via a web form | I can quickly check if a connection is malicious          |
| US-02 | Security Analyst       | View a live dashboard of attack statistics     | I can understand the overall threat landscape at a glance |
| US-03 | Security Analyst       | Browse a log of all past detections            | I can audit and investigate past alerts                   |
| US-04 | Developer / DevOps     | Call a REST API with traffic features          | I can automate detection in my monitoring pipeline        |
| US-05 | Student / Demonstrator | Click demo buttons for each attack type        | I can showcase the system without real network data       |

## 5.2 Solution Architecture

The system follows an MVC-style web application architecture with an integrated ML inference layer:

| Layer        | Component     | Technology                   | Responsibility                                        |
|--------------|---------------|------------------------------|-------------------------------------------------------|
| Presentation | Web Interface | HTML5, Bootstrap 5, Chart.js | Responsive UI, forms, dashboard, alert table          |
| Application  | Flask Router  | Python 3, Flask 3.1          | Request routing, session handling, template rendering |
| Inference    | ML Pipeline   | scikit-learn, NumPy          | Feature scaling + Random Forest classification        |
| Storage      | Alert Store   | In-memory Python list        | Logging predictions with timestamp and metadata       |
| Integration  | REST API      | Flask JSON, HTTP POST        | Programmatic access for external systems              |
| Model        | Pickle Files  | .pkl (joblib/pickle)         | Persisted StandardScaler and Random Forest model      |

## 6. PROJECT PLANNING & SCHEDULING

### 6.1 Technical Architecture

| Category          | Technology     | Version    | Purpose                                   |
|-------------------|----------------|------------|-------------------------------------------|
| Backend Runtime   | Python         | 3.10+      | Core programming language                 |
| Web Framework     | Flask          | 3.1        | HTTP routing and templating               |
| Machine Learning  | scikit-learn   | 1.4+       | Random Forest Classifier + StandardScaler |
| Data Processing   | NumPy / Pandas | 1.26 / 2.x | Array operations and dataset handling     |
| Frontend CSS      | Bootstrap      | 5.3        | Responsive grid and UI components         |
| Frontend Charts   | Chart.js       | 4.x        | Doughnut chart on dashboard               |
| Model Persistence | Pickle (.pkl)  | stdlib     | Save and load trained model artifacts     |
| Dataset           | NSL-KDD        | —          | Network intrusion labeled training data   |
| Version Control   | Git & GitHub   | 2.x        | Source code management and collaboration  |

### 6.2 Sprint Planning & Estimation

| Sprint   | Goal                     | Tasks                                                            | Estimate |
|----------|--------------------------|------------------------------------------------------------------|----------|
| Sprint 1 | Environment & Data Setup | Install dependencies, download NSL-KDD, EDA, feature selection   | 3 days   |
| Sprint 2 | Model Development        | Preprocessing, model training, evaluation, save .pkl files       | 3 days   |
| Sprint 3 | Flask Backend            | Create app.py, define routes (/predict, /alerts, /api/predict)   | 3 days   |
| Sprint 4 | Frontend Templates       | Build home, dashboard, predict, alerts, about HTML templates     | 3 days   |
| Sprint 5 | Integration & Testing    | Connect ML pipeline to routes, validate API, performance testing | 2 days   |
| Sprint 6 | Documentation & Demo     | Write README, project report, prepare demo, GitHub push          | 2 days   |



### 6.3 Sprint Delivery Schedule

| Sprint   | Week     | Deliverable                                  | Status    |
|----------|----------|----------------------------------------------|-----------|
| Sprint 1 | Week 1   | Cleaned dataset + feature selection notebook | Completed |
| Sprint 2 | Week 1–2 | ids_model.pkl + scaler.pkl + accuracy report | Completed |
| Sprint 3 | Week 2   | app.py with all routes functional            | Completed |
| Sprint 4 | Week 2–3 | All 5 HTML templates rendered correctly      | Completed |
| Sprint 5 | Week 3   | End-to-end test results + API verification   | Completed |
| Sprint 6 | Week 4   | GitHub repo + README + Project Documentation | Completed |

## 7. CODING & SOLUTIONING

### 7.1 Feature 1: ML Model Training (train\_model.py)

This feature encompasses the complete machine learning pipeline — from raw data ingestion through feature engineering, model training, evaluation, and artifact persistence.

#### Key Implementation Steps:

- Load the NSL-KDD dataset using Pandas and assign the 10 selected feature columns.
- Encode the label column into numeric classes: 0=Normal, 1=DoS, 2=Probe, 3=R2L, 4=U2R.
- Apply StandardScaler to normalize feature values to zero mean and unit variance.
- Train a RandomForestClassifier with 100 decision trees using an 80/20 train-test split.
- Evaluate using accuracy score, classification report, and confusion matrix.
- Persist both the trained model and scaler as .pkl files using joblib.

#### Code Snippet — train\_model.py:

```
import pandas as pd from sklearn.ensemble import RandomForestClassifier from sklearn.preprocessing import StandardScaler,
LabelEncoder from sklearn.model_selection import train_test_split from sklearn.metrics import accuracy_score,
classification_report import joblib FEATURES = ['duration','src_bytes','dst_bytes','land', 'wrong_fragment','urgent','hot',
'num_failed_logins','logged_in','num_compromised'] # Load and prepare data df = pd.read_csv('data/ns_l_kdd_train.csv') X =
df[FEATURES] y = LabelEncoder().fit_transform(df['label']) # Scale features scaler = StandardScaler() X_scaled =
scaler.fit_transform(X) # Train X_train, X_test, y_train, y_test = train_test_split(X_scaled, y, test_size=0.2) model =
RandomForestClassifier(n_estimators=100, random_state=42) model.fit(X_train, y_train) print(f"Accuracy:
{accuracy_score(y_test, model.predict(X_test)):.4f}") # Save artifacts joblib.dump(model, 'models/ids_model.pkl')
joblib.dump(scaler, 'models/scaler.pkl')
```

### 7.2 Feature 2: Flask Web Application & REST API (app.py)

The Flask application serves as the central hub, connecting the trained ML model to the web interface and REST API. It defines six routes, manages an in-memory alert store, and handles both form-based and JSON-based prediction requests.

| Route        | Method    | Handler Function | Description                                                               |
|--------------|-----------|------------------|---------------------------------------------------------------------------|
| /            | GET       | home()           | Renders landing page with feature overview and attack type guide          |
| /dashboard   | GET       | dashboard()      | Renders live stats dashboard with Chart.js doughnut chart                 |
| /predict     | GET, POST | predict()        | Prediction form; on POST scales features and returns result card          |
| /alerts      | GET       | alerts()         | Renders paginated table of all logged predictions                         |
| /about       | GET       | about()          | Project details, tech stack overview, and API documentation               |
| /api/predict | POST      | api_predict()    | JSON REST endpoint — returns label, confidence, is_attack flag, timestamp |

Code Snippet — Core Prediction Logic in app.py:

```
@app.route('/api/predict', methods=['POST']) def api_predict(): data = request.get_json() features =
np.array(data['features']).reshape(1, -1) scaled = scaler.transform(features) pred = model.predict(scaled)[0] prob =
model.predict_proba(scaled)[0] label_map = {0:'Normal',1:'DoS Attack',2:'Probe',3:'R2L Attack',4:'U2R Attack'} label =
label_map[pred] confidence = round(float(prob[pred]) * 100, 1) alert = { 'label': label, 'confidence': confidence,
'is_attack': pred != 0, 'timestamp': datetime.now().isoformat() } alerts_store.append(alert) return jsonify(alert)
```

7.3 Database Schema (In-Memory Alert Store)

The system uses an in-memory Python list as its alert store. Each entry follows a consistent schema. For production deployment, this can be replaced with a SQLite or PostgreSQL table using the same schema.

| Field      | Type       | Example Value                       | Description                                        |
|------------|------------|-------------------------------------|----------------------------------------------------|
| label      | String     | "DoS Attack"                        | Human-readable attack category name                |
| confidence | Float      | 92.3                                | Model confidence as percentage (0–100)             |
| is_attack  | Boolean    | true                                | True if predicted class is not Normal (class != 0) |
| timestamp  | ISO String | "2024-10-15T14:32:00"               | UTC timestamp of prediction event                  |
| features   | Array[10]  | [0, 490000, 0, 0, 3, 0, 0, 0, 0, 0] | Original 10 input features for audit trail         |

## 8. PERFORMANCE TESTING

### 8.1 Performance Metrics

The following performance metrics were recorded during testing on a standard development machine (Intel Core i5, 8GB RAM, Python 3.10):

| Metric                      | Value         | Measurement Method                                 |
|-----------------------------|---------------|----------------------------------------------------|
| Model Accuracy (Test Set)   | 96–100%       | sklearn accuracy_score on 20% holdout              |
| Precision (Weighted Avg)    | 0.97          | classification_report — weighted average           |
| Recall (Weighted Avg)       | 0.96          | classification_report — weighted average           |
| F1-Score (Weighted Avg)     | 0.96          | classification_report — weighted average           |
| API Response Time (avg)     | < 80ms        | curl -w %{time_total} over 100 sequential requests |
| Web Form Prediction Time    | < 200ms       | Browser DevTools Network tab (including render)    |
| Model Load Time (startup)   | < 1.5 seconds | Time from app.py start to first request served     |
| Memory Usage (steady state) | ~120 MB       | Measured via psutil during 1,000 request load test |

| Attack Class | Precision | Recall | F1-Score | Support  |
|--------------|-----------|--------|----------|----------|
| Normal       | 0.99      | 0.98   | 0.98     | High     |
| DoS Attack   | 0.97      | 0.99   | 0.98     | High     |
| Probe        | 0.96      | 0.95   | 0.95     | Medium   |
| R2L Attack   | 0.94      | 0.93   | 0.93     | Low      |
| U2R Attack   | 0.92      | 0.90   | 0.91     | Very Low |

## 9. RESULTS

### 9.1 Output Screenshots

The following sections describe the key screens of the deployed application.

#### Home Page (/)

The landing page presents an overview of the system with a hero section, a feature highlights grid (Real-time Detection, ML-Powered, REST API, Alert Logging), and an attack category reference table. Navigation links direct users to Dashboard, Predict, and Alerts sections.

*[Screenshot: Home Page — Hero section with navigation, feature cards, and attack type table]*

#### Dashboard (/dashboard)

The dashboard displays real-time statistics including total predictions, attack count, and normal traffic count. A Chart.js doughnut chart visualizes the distribution of all five traffic categories. The chart updates dynamically as new predictions are logged.

*[Screenshot: Dashboard — Live stat cards and doughnut chart showing traffic distribution]*

#### Prediction Form (/predict)

The prediction page presents a 10-field input form mapped to the NSL-KDD features. Demo shortcut buttons (Normal, DoS, Probe, R2L, U2R) auto-fill the form with representative values. Upon submission, a result card displays the predicted label, a color-coded threat badge, and a confidence percentage.

*[Screenshot: Predict Page — 10-feature form with demo buttons and prediction result card]*

#### Alert Log (/alerts)

The alerts page renders a full tabular log of every prediction made since application start. Each row displays the sequence number, timestamp, predicted label, a color-coded badge, and a visual confidence progress bar. This provides a complete audit trail for security review.

*[Screenshot: Alerts Page — Table of logged detections with timestamps, labels, and confidence bars]*

## 10. ADVANTAGES & DISADVANTAGES

| Advantages                                                        | Disadvantages                                                            |
|-------------------------------------------------------------------|--------------------------------------------------------------------------|
| High accuracy (96–100%) across all five attack categories         | Alert store is in-memory; data is lost on server restart                 |
| No manual rule updates required — model learns from data          | Model currently uses only 10 of 41 NSL-KDD features, limiting coverage   |
| User-friendly web interface accessible without security expertise | Does not perform live packet capture; requires manual feature input      |
| REST API enables integration with existing SOC tooling            | Susceptible to adversarial feature manipulation (evasion attacks)        |
| Fast inference time (< 80ms per API call)                         | Trained on static NSL-KDD data; may not generalize to new attack types   |
| Fully open-source stack with no licensing costs                   | No authentication or access control on web interface or API              |
| Confidence scores add interpretability beyond binary decisions    | Single-server architecture not suitable for high-availability deployment |

## 11. CONCLUSION

The AI-Enhanced Intrusion Detection System successfully demonstrates the practical application of machine learning in the field of network cybersecurity. By training a Random Forest Classifier on the well-established NSL-KDD dataset and integrating it into a full-stack Flask web application, the project delivers a functional, accessible, and extensible tool for real-time network intrusion detection.

The system correctly identifies all five classes of network traffic — Normal, DoS, Probe, R2L, and U2R — with accuracy consistently exceeding 96%. The inclusion of a live dashboard powered by Chart.js, a comprehensive alert logging system, interactive demo simulation buttons, and a documented REST API positions this project as a complete, deployable solution suitable for academic demonstration, internship showcases, and small-scale network monitoring scenarios.

This project illustrates how modern Python tooling (Flask, scikit-learn, NumPy, Pandas) can be combined to create production-grade security applications that go beyond theoretical models. The architecture is intentionally modular, enabling straightforward replacement of the ML model, addition of new features, and integration with persistent databases or external SIEM platforms.

## 12. FUTURE SCOPE

| Enhancement               | Description                                                                                                           | Priority |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------|----------|
| Live Packet Capture       | Integrate Scapy or PyShark to capture and analyze live network traffic, eliminating the need for manual feature entry | High     |
| Persistent Database       | Replace the in-memory alert store with PostgreSQL or SQLite for durable logging and historical analysis               | High     |
| User Authentication       | Add Flask-Login based authentication to secure the web interface and API with role-based access control               | High     |
| Model Retraining Pipeline | Build an automated pipeline to retrain the model on newly collected traffic data at scheduled intervals               | Medium   |
| Full 41-Feature Support   | Extend the feature set to all 41 NSL-KDD features to improve detection coverage and reduce false negatives            | Medium   |
| Containerization          | Dockerize the application with docker-compose to enable one-command deployment across any environment                 | Medium   |
| Email / SMS Alerting      | Send automated notifications via email (SMTP) or SMS (Twilio) when high-confidence attacks are detected               | Medium   |
| Deep Learning Comparison  | Implement and benchmark LSTM or CNN-based models against the current Random Forest baseline                           | Low      |
| SIEM Integration          | Develop a connector to forward alerts to enterprise SIEM platforms such as Splunk or IBM QRadar                       | Low      |



### 13. APPENDIX

#### Source Code — File Structure

```
AI-Enhanced-Intrusion-Detection-System/ |— app.py           # Flask application entry point |— train_model.py      #
ML training pipeline |— requirements.txt      # Python dependencies |— README.md          # Project documentation
|— models/ |— ids_model.pkl                  # Trained Random Forest model |— scaler.pkl        # Fitted StandardScaler
|— templates/ |— home.html                  # Landing page |— dashboard.html    # Live stats + Chart.js |—
predict.html      # 10-feature prediction form |— alerts.html      # Alert log table |— about.html    # About +
API docs |— static/css/ |— style.css        # Custom stylesheet |— Documentation/ |—
Project_Documentation.docx
```

#### GitHub & Project Demo Link

| Resource          | URL / Details                                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| GitHub Repository | <a href="https://github.com/sourabhpatil/AI-Enhanced-Intrusion-Detection-System">https://github.com/sourabhpatil/AI-Enhanced-Intrusion-Detection-System</a> |
| Project Demo      | Run locally: python app.py → http://localhost:5000                                                                                                          |
| API Endpoint      | POST http://localhost:5000/api/predict (JSON body: {"features": [...]})                                                                                     |
| Training Data     | NSL-KDD Dataset — <a href="https://www.unb.ca/cic/datasets/nsl.html">https://www.unb.ca/cic/datasets/nsl.html</a>                                           |
| Submitted By      | Sourabh Bhimrao Patil   SmartInternz Externship Program 2026                                                                                                |

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